Batch: B-2	**Roll No.: 16010124107**
Experiment No.2	

Title: Design of EER diagram and Mapping to Relational Model

Objective: Design the Enhanced Entity-Relationship (EER) diagram and its relational mapping

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Expected Outcome of Experiment:

CO1: Comprehend the Characteristics of Relational Database Management Systems.
CO2: Create Relational Database Designs Based on Entity-Relationship Models.

Books/ Journals/ Websites referred:

1. G. K. Gupta :"Database Management Systems", McGraw – Hill
2. Korth, Slberchatz, Sudarshan : "Database Systems Concept", 6th Edition , McGraw Hill
3. Elmasri and Navathe, "Fundamentals of Database Systems", 5thEdition, PEARSON Education.

Pre Lab/ Prior Concepts:

The ER data model was developed to facilitate the database design by allowing specification of an enterprise schema that represents the overall logical structure of the database. The ER model is one of the several data models. The semantic aspect of the model lies in its representation of the meaning of the data. The ER model is very useful many database design tools drawn on concepts from the ER model. The ER model employs 3 basic notations: entity set, relationship set and

attributes.

Extended Entity Relationship Diagram:

The EER model includes all of the concepts introduced by the ER model. Additionally it includes the concepts of a subclass and superclass (Is-a), along with the concepts of specialization and generalization. Furthermore, it introduces the concept of a union type or category, which is used to represent a collection of objects that is the union of objects of different entity types. EER model also includes EER diagrams that are conceptual models that accurately represent the requirements of complex databases.

Example Case Study: List the data requirements for the database of the company which keeps track of the company employee, department and projects. The database designers provide the following description

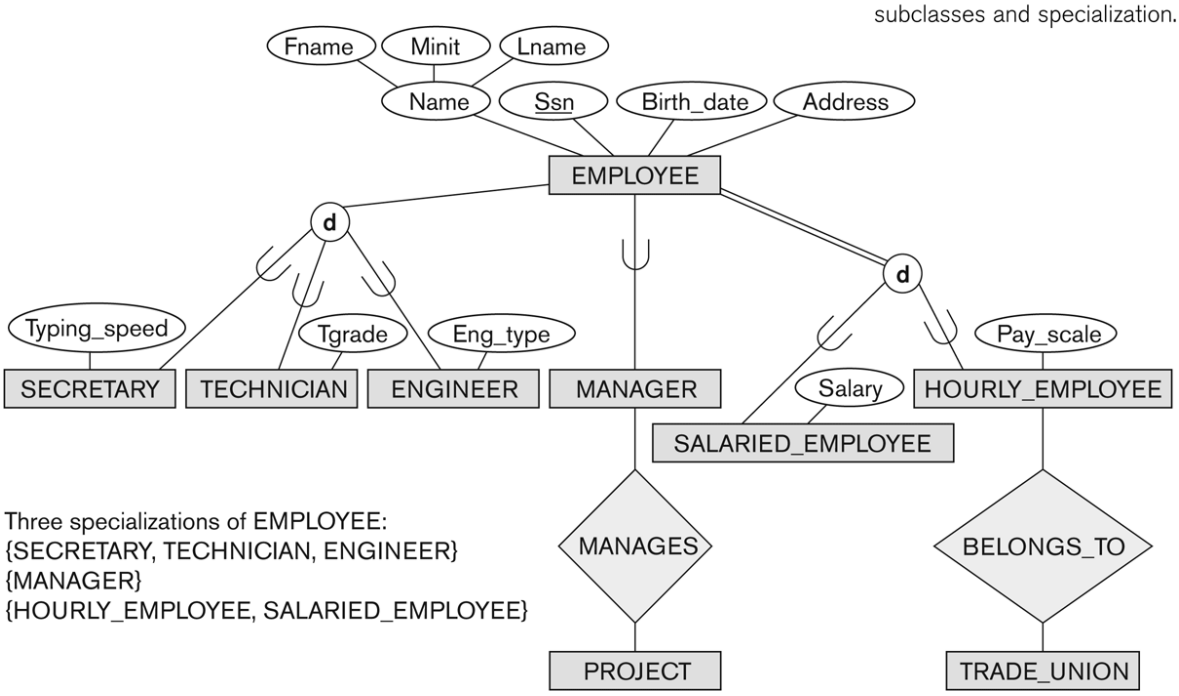
Procedure for doing the EER diagram experiment

EER stands for Enhanced ER or Extended ER
EER Model Concepts
Includes all modeling concepts of basic ER
Additional concepts:

- subclasses/superclasses
- specialization/generalization
- categories (UNION types)
- attribute and relationship inheritance

These are fundamental to conceptual modeling
The additional EER concepts are used to model applications more completely and more accurately
EER includes some object-oriented concepts, such as inheritance

Figure 4.1
EER diagram notation to represent subclasses and specialization.



Generalization

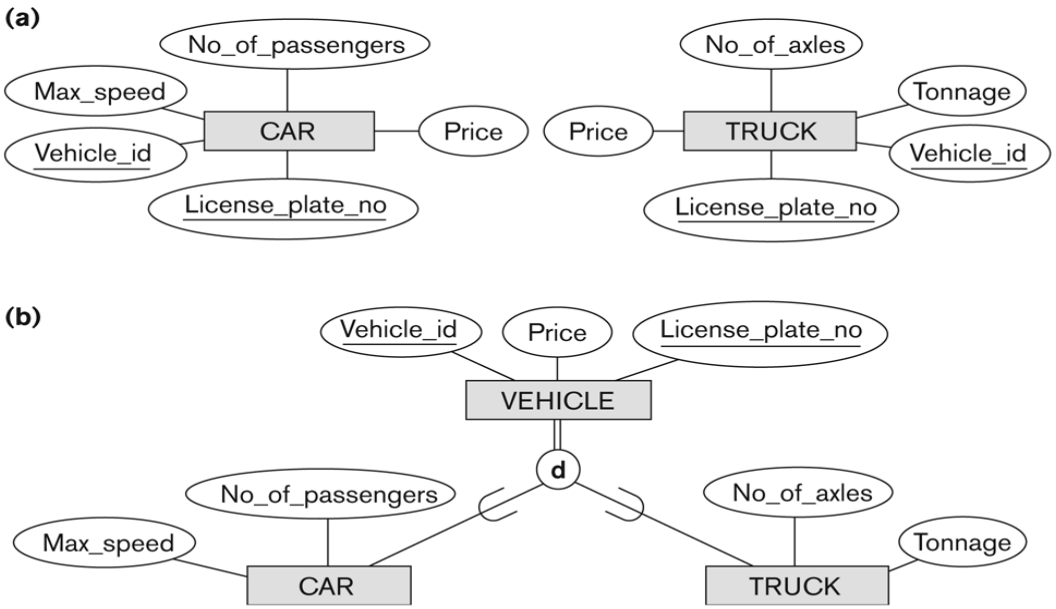


Figure 4.3
Generalization. (a) Two entity types, CAR and TRUCK.
(b) Generalizing CAR and TRUCK into the superclass VEHICLE.

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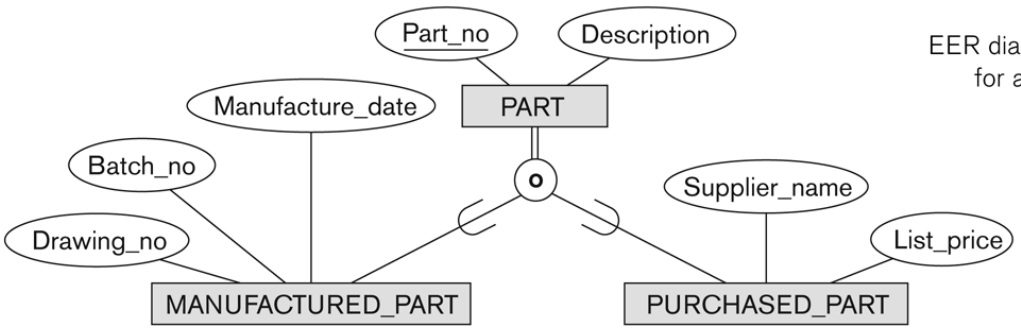
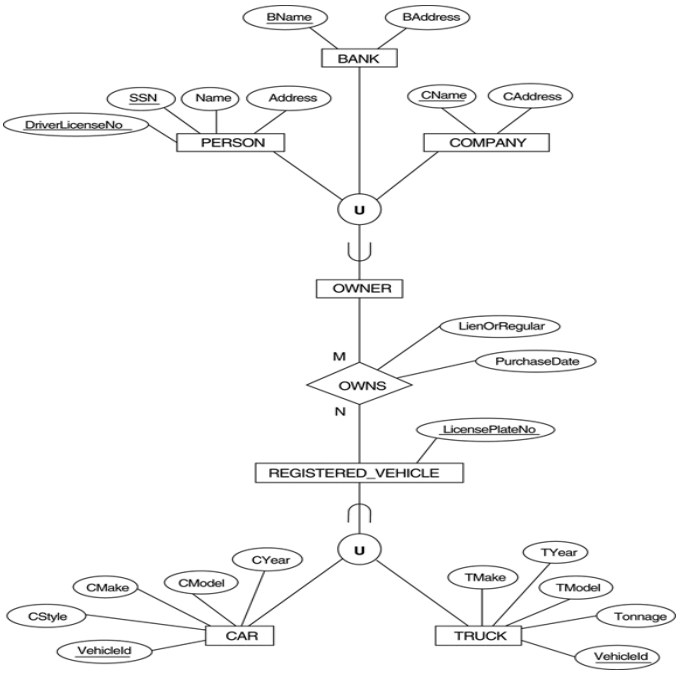
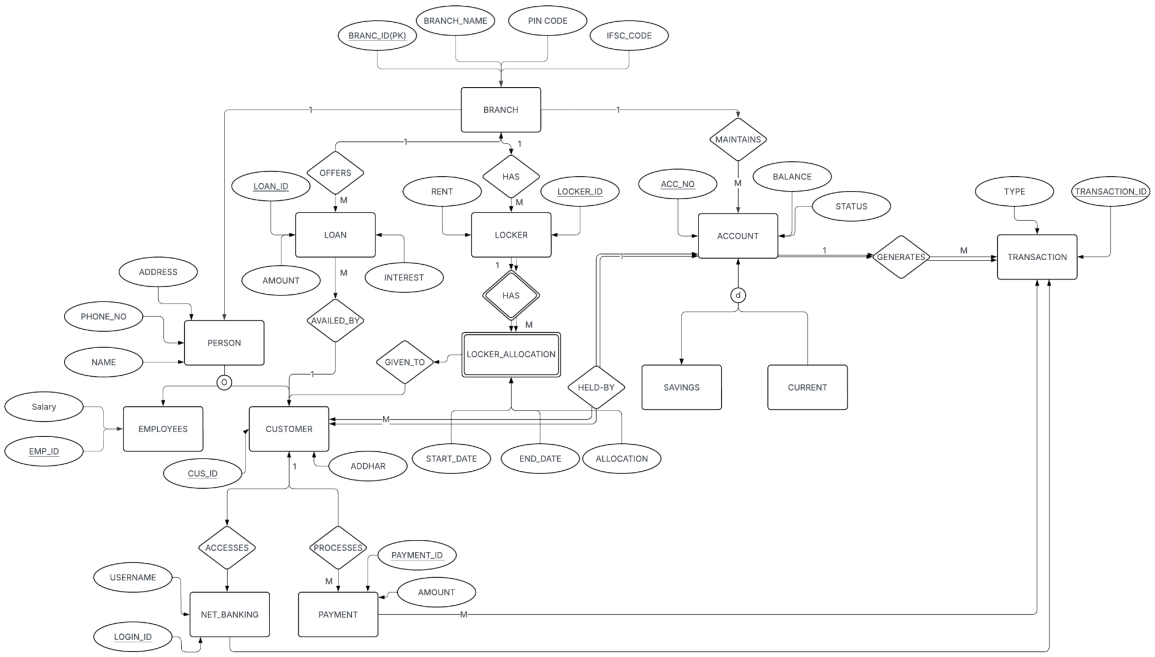


Figure 4.5
EER diagram notation for an overlapping (nondisjoint) specialization.

C: Categories and Union



OUTPUT:

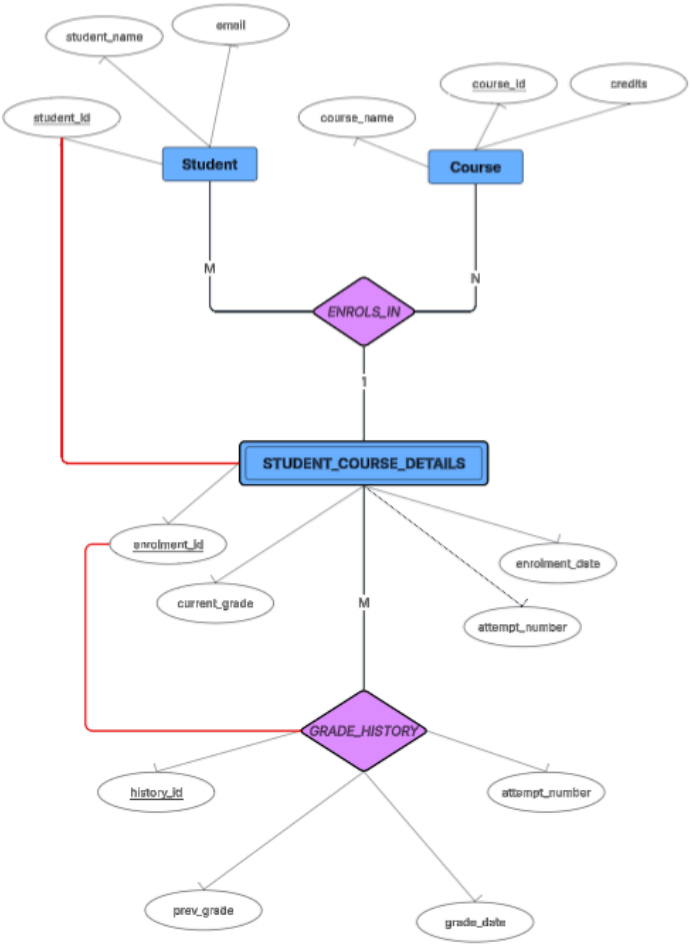


Conclusion :

Enhanced EER model improves banking design using specialization, generalization, and weak entities, ensuring data consistency, reducing redundancy, enforcing integrity, and enabling efficient relational database implementation.

Post Lab Descriptive Questions

1. In the Academic database a Grade is issued to each STUDENT for each COURSE taken and stored in the STUDENT COURSE DETAIL entity. A STUDENT may decide to re-take a COURSE to better their GRADE. The administration would like to keep a record of the old/previous Grade as well as the new Grade. Show ER diagram to include historical Grades if the students should have them.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